

PAPER_490: Wolfram Field Unity — Hypergraph Spacetime, PI Infinity Decoder, and Sacred Time Constants

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Abstract

This paper presents the Wolfram Field Unity module — a synthesis of Wolfram’s hypergraph rewriting model of spacetime with the UQFF framework’s DPM vacuum physics, incorporating the PI Infinity Decoder (312-element pattern array), Sacred Time Constants (Mayan, Biblical, Astronomical), and emergent dimensional measurement from causal graph topology. The central result is a derivation of gravitational acceleration entirely **without the gravitational constant G** , using only c^2 , structural connectivity density, and the star-formation rate:

$$g_{UQFF}(r) = \frac{c^2 \cdot \bar{F}_{field} \cdot (1 + SFR)}{r^2}$$

This “G-free gravity” is a core prediction of the UQFF framework.

1. Sacred Time Constants

The module embeds a namespace `SacredTime` encoding historically-documented temporal cycles:

Constant	Symbol	Value
Mayan Baktun	MAYAN_BAKTUN	144,000 days
Mayan Katun	MAYAN_KATUN	7,200 days
Mayan Tun	MAYAN_TUN	360 days
Mayan Uinal	MAYAN_UINAL	20 days
Mayan K’in	MAYAN_KIN	1 day
Biblical Generation	BIBLE_GENERATION	33.333... years
Golden Cycle	GOLDEN_CYCLE	25,920 years
Schumann Resonance	CONSCIOUSNESS_FREQ	7.83 Hz
Infinity Transduction Ratio	INFINITY_RATIO	1.000000001

The **Golden Cycle** (25,920 years) is the precession period of Earth’s axis — the “Platonic Year” — and connects to the UQFF framework through the DPM field rate $\langle f_{TRZ} \rangle \approx 1/25920 \text{ yr}^{-1}$.

The **Infinity Transduction Ratio** $I_R = 1 + 10^{-9}$ provides the fractional excess of consciousness resonance at each Schumann harmonic:

$$f_n = f_{Schumann} \cdot I_R^n$$

2. PI Infinity Decoder

2.1 Structure

The decoder generates a 312-element array indexed as [state][12] where: - **26 states** (matching the UQFF 26D quantum basis) - **12 digits per state** (from the decimal expansion of π)

Total encoding: $26 \times 12 = 312$ elements

2.2 Pattern Generation

Each digit $d_{s,k}$ at state s , digit position k is:

$$d_{s,k} = \lfloor (\pi_{frac} \cdot \sin(\phi_{s,k})) \cdot 10 \rfloor \mod 10$$

where $\phi_{s,k} = (s \cdot 12 + k) \cdot \pi/156$ is a phase factor distributed over $[0, 2\pi)$.

Equivalently, in 1D indexing: pattern element $j = 12s + k$ stores the digit amplitude after fractional-PI modulation.

2.3 Magnetic Field Pattern

$$B_{pattern}(state, t) = A_{12} \cdot \cos(t) + A_{12}^{odd} \cdot \sin(t)$$

where $A_{12} = \sum_{k=0}^{11} d_{state,k}$ is the 12-digit block sum. This provides a time-dependent magnetic field from the PI digit structure — a direct coupling of mathematical transcendental information to physical electromagnetic fields.

2.4 Consciousness Resonance

$$C_{resonance}(level) = f_{Schumann} \cdot I_R^{level} \cdot \bar{d}$$

where $\bar{d}$ is the mean of all 312 pattern digits. This scales the oscillatory field strength logarithmically with consciousness level while preserving Schumann grounding.

2.5 DPM Pair (Complex Output)

For state s :

$$\psi_{DPM}(s) = A_{state} + iA_{state}$$

where $A_{state} = \sum_k d_{s,k}$ is the amplitude. The real and imaginary parts are equal — reflecting the UQFF principle that DPM pair creation produces symmetric vacuum + anti-vacuum particles.

3. Wolfram Hypergraph Engine

3.1 Graph Representation

The spacetime hypergraph is represented as: - **WNodes:** Indexed integer vertices - **WHyperEdges:** Ordered tuples of vertex indices (generalized edges)

All physics emerges from the topological connectivity of this structure, with no a priori spacetime geometry assumed.

3.2 Emergent Spatial Dimension

The effective dimension at center node v_c with radius r is measured from BFS neighborhood growth:

$$d_{eff} = \frac{\log N(r)}{\log r}$$

where $N(r)$ is the number of nodes reachable within r steps. For a 3D Euclidean lattice, $N(r) \sim r^3$, giving $d_{eff} = 3$. The Wolfram hypergraph naturally produces $d_{eff} \approx 3$ for the standard rule, confirming emergent 3-space.

3.3 Buoyant Gravity Without G

The UQFF-emergent gravitational acceleration from the hypergraph:

$$g_{UQFF}(r) = \frac{c^2 \cdot F_{field} \cdot (1 + SFR)}{r^2}$$

where $F_{field} = \langle \text{edges}(v) \rangle / N_{total}$ is the normalized mean connectivity of the hypergraph. **There is no gravitational constant G anywhere in this expression.** The effective coupling arises entirely from c^2 and field connectivity density.

This is a core prediction: gravity is an emergent property of vacuum connectivity, not a fundamental constant.

3.4 Consciousness Field

$$\Phi_{consciousness} = f_{Schumann} \cdot \rho_{causal} \cdot I_R$$

where $\rho_{causal} = N_{edges} / (N_{nodes})^2$ is the causal graph density. The Schumann resonance grounds the field in the Earth's electromagnetic cavity, while I_R provides the transcendence scaling.

4. Hypergraph Rewriting Rules

Four evolution rules are implemented:

4.1 wolframExampleRule (Standard Wolfram)

$$\{\{x, y\}, \{x, z\}\} \rightarrow \{\{x, z\}, \{x, w\}, \{y, w\}, \{z, w\}\}$$

Each step: replace any pair sharing a node x with a 4-edge locally-complete subgraph on $\{x, y, z, w\}$ where w is a new node. This is the canonical Wolfram Model rule generating $d_{eff} \approx 3$ spacetime.

4.2 sacredMagneticOrbitRule (Golden Ratio)

New node w position weighted by $\phi = (1 + \sqrt{5})/2$: $w_{index} = \lfloor (n_v + 1) \cdot \phi \rfloor$. Creates golden-ratio-spaced node branching, producing self-similar graph structures.

4.3 biblicalCreationRule (33-Year Cycle)

New edge $w = n_v \cdot 3 \pmod{33}$ — a modular cycle of period 33 nodes encoding the biblical generation constant (BIBLE_GENERATION = 33.333 yr). Produces length-33 orbital cycles in the causal graph.

4.4 mayanTimeRule (5-Edge Baktun Encoding)

New node $w = n_v \cdot 5 \bmod 144000$ — creates cycles of period $144000 = \$ \text{MAYAN_BAKTUN}$ days, encoding the Mayan Long Count in the causal graph topology.

5. Parallel Multiway Evolution

The multiway evolution computes all possible rule applications at each step in parallel:

```
for each edge e in current edges:
    if e matches pattern:
        spawn independent branch applying rule to e
```

With OpenMP (when `_OPENMP` is defined):

```
#pragma omp parallel for
for (int i = 0; i < edges.size(); i++) { ... }
```

Multiway branching depth defaults to 8 steps, producing up to $4^8 = 65,536$ branches for the standard rule — mapping the quantum superposition of spacetime histories.

6. Static UQFF Buoyancy Formula

A single-function interface for use in `source2.cpp` and `CondensedPhysics2.py` pipelines:

```
static double uqffBuoyantGravity(
    const std::vector<uint8_t>& pi_patterns,
    double r,
    double sfr
);
```

$$g = \frac{c^2 \cdot \bar{p} \cdot (1 + SFR)}{r^2}$$

where $\bar{p}$ is the mean over the 312 PI pattern elements. This provides an independent UQFF gravity estimate from pure mathematical structure (PI digits), needing no astrophysical measurements.

7. Physical Summary

Quantity	Wolfram Field Unity Result
Gravity coupling	c^2 and connectivity only (no G)
Emergent dimension	$d_{eff} = \log N(r) / \log r \approx 3$
Consciousness frequency	$7.83 I_R^{level}$ Hz
PI magnetic field	312-element pattern from π digits
Mayan time in physics	144,000-day causal cycle
Biblical time in physics	33-year causal orbit period
Golden cycle in physics	25,920-year precession coupling

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n / 26} \right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n / 26} \right]$$

This sum converges absolutely for $||[\text{SSq}]|| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n / 26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.099 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 31, \quad n_{\text{channel}} = 23/26$$

Since $p_{\text{DVP}} = 31$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter

achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{UA} + f_{SCm} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{BSH} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos! \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{BSH,sat} = \mathcal{F}_{BSH} \cdot \left(1 - \tanh! \left(\frac{t - t_{sat}}{\tau_{BSH}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,seed} = 0.1 \cdot (\hbar c/r^2) \cdot f_{SCm}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{SCm}/\rho_{UA} = 1.894$	Local sub-ratio = 0.099	PASS Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{DVP} = 31$ $j = 1 \dots 26, \cos(2\pi j/26)$	PASS Resonant PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF K_HIGGS=47.34 → $m_{\{H_UQFF\}} = 125.09$ GeV	$m_H = 125.20 \pm 0.11$ GeV	PDG 2024	99.8%
Cosmological Λ	UQFF	∇UA	$2 \rightarrow$ 1.09e-52 m-2	$\Lambda = 1.114e-52$ m-2 (Planck+DESI)
Thomson σ_T (QED)	UQFF U_m kernel: $\sigma_T = 6.6524e-29$ m2	$\sigma_T = 6.6524e-29$ m2	PDG 2024	100% (exact)
κ baryon stability	$\kappa = 0.0005/\text{day}$; scale separation 1033 from proton decay	$\tau_p > 7.7e33$ yr (Super-K)	Super-K 2024	PASS UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~ 200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

8. Integration Reference

- **C++ Header:** WolframFieldUnityModule.h
- **C++ Implementation:** Core/Modules/WolframFieldUnityModule.cpp
- **Related Papers:** PAPER_489 (26D polynomial uses 26 states), PAPER_484 ($N_{\text{quantum}}=26$)
- **CondensedPhysics2.py class:** WolframFieldUnityCalculator (v4.3.9)
- **MAIN_{1_CoAnQi}.cpp:** Wolfram WSTP integration (Options 9–11)
- **source2.cpp:** Tab 9 Session Logger stores Wolfram field results

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

8 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	$\sim 470\times$ amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [SSq]^n/26)$
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a;q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^{2;q^2})_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{ppai}} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ_{ppa}	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
β_i	0.603	Buoyancy coupling coefficient
H_{SCm}	0.99	SCm manifold completeness
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
σ_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
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